Sensor systems

Neck EMG-Based Estimation of Upper-Limb Muscle Tension

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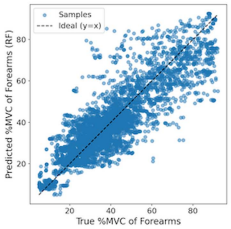
Abstract—Monitoring muscle tension unobtrusively is important for sports and music performance. Conventional approaches estimate bilateral arm exertion by placing electromyography (EMG) sensors on the arms; however, in sports the arms may strike players, risking sensor detachment or damage, and in artistic settings visible arm-mounted sensors may be socially unacceptable. To address these issues, I propose a method that estimates bilateral arm exertion using a single EMG sensor on the posterior neck. The approach leverages the kinetic-chain principle: when the arms tense, cervical muscles coactivate to stabilize posture. Using cervical EMG and machine-learning models, I classify arm tension and regress percent maximum voluntary contraction (%MVC) across both arms. I evaluated the method across five tasks including grip strength, golf putting, and piano playing. The method achieved a mean binary-classification accuracy of 76%. For regression of mean arm %MVC, it yielded an average RMSE of 10% and R^2 of 0.72.



Index Terms—Sensor systems, electromyography (EMG), muscle, neck, wearable.

I. INTRODUCTION

Muscle fatigue and excessive contraction (overexertion) compromise both performance and safety and should therefore be avoided [1], especially in sports and musical performance. For example, studies of golf [2] and piano performance [3] report that undue muscular tension degrades performance and increases injury risk. Conventional upper-limb monitoring relies on attaching multiple surface electromyography (EMG) electrodes to the arms. For example, in piano keystroke studies, EMG sensors were placed on six muscles of the right arm [4]: posterior deltoid, anterior deltoid, triceps brachii (lateral head), biceps brachii, flexor digitorum superficialis, and extensor digitorum communis. In the golf putting study [5], EMG sensors were placed on five muscles of each arm: flexor digitorum superficialis, extensor digitorum, biceps brachii, triceps brachii, and deltoid. While elevated arm EMG can indicate overexertion, fastening several sensors to the forearms may interfere with natural movement during performance [6] and discourage adoption [7].

To address these issues, I explore a novel sensor placement on the posterior neck (back of the neck). Although the neck is not yet a mainstream site for wearable sensors, it has significant untapped potential. Compared with the arms, the posterior neck experiences far less contact with players or equipment in sport, reducing the risk of sensor detachment or damage. From a social-acceptability standpoint, a posterior-neck sensor is also less conspicuous than arm-mounted sensors, making it more acceptable in artistic contexts. Indeed, prior work has mounted IMUs at this location for posture monitoring [8], and even in high-contact sports such as rugby, GPS units and accelerometers are routinely worn on the back of the neck [9]. Building on these precedents, I propose mounting an EMG sensor on the posterior neck.

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The proposed method estimates upper-limb tension using a single EMG sensor on the posterior neck, based on the hypothesis that cervical musculature coactivates when the upper limbs exert force. This premise is grounded in biomechanical findings on kinetic chains [10] and muscle synergies [11]. In sports and musical performance, neck activation is thought to stabilize the head, enable efficient force transmission along the kinetic chain, and operate as part of a coordinated synergy that supports upper-limb movement. Prior studies show that arm actions recruit cervical stabilizers during running [12], tapping [13], and isometric tasks [14]. Unlike traditional methods that directly measure local muscle load [15], the proposed method indirectly estimates upper-limb overexertion from neck EMG.

This article presents a user study investigating the feasibility of estimating upper-limb overexertion with a single EMG sensor positioned over the upper trapezius on the posterior neck. Eighteen participants performed five tasks including grip strength, golf putting, and piano playing under two effort conditions (Forceful and Relaxed). EMG signals were recorded from the neck and both forearms, and muscle activity was expressed as percent maximum voluntary contraction (%MVC). For each task, I collected training and test sets, trained classification and regression models, and evaluated binary detection of upper-limb tension and regression of mean arm %MVC.

II. MATERIALS AND METHODS

I conducted a user study to assess whether cervical EMG recorded from a single sensor on the posterior neck can serve as a reliable basis for estimating upper-limb muscle tension. Participants performed five tasks under two effort instructions. These tasks span distinct movement categories (static-isometric, whole-body, and fine-motor). In previous studies, both golf [5] and piano [4] have been analyzed using EMG sensors; therefore, I selected their basic skills as tasks in this study.

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The study protocol was approved by my institutional ethical review board (approval number: Gitairin-2024-02, date of approval: 24 July 2024).

Eighteen adults (9 females, 9 males; ages 20 s–40 s; $M = 33.6$ years, $SD = 8.33$) participated. All were healthy and right-handed. Written informed consent was obtained from each participant. None of the participants had any formal training in golf or piano.

A. Apparatus

Surface EMG (sEMG) was recorded with the Trigno Wireless System (Delsys, Inc.). Sensors were placed over the upper trapezius on the posterior neck and over the extensor digitorum on both forearms. The upper trapezius contributes to scapular elevation and postural stabilization, and the extensor digitorum cocontracts with the flexor digitorum superficialis during forceful exertion. Raw EMG was acquired using Trigno Discover (v1.6.2) at 1259 Hz, the device's default sampling rate.

B. Procedure

Prior to the experimental tasks, I recorded maximum voluntary contractions (MVCs) for each muscle to normalize EMG signals. The MVC of the upper trapezius was recorded during a maximal shoulder shrug while participants supported the posterior neck with their hands and maximally extended the neck. The MVC of the extensor digitorum was recorded during maximal handgrip. For each muscle, data were recorded for 3 s.

After calibration, participants completed the five tasks in the fixed order described below. Each task was performed once under the Forceful condition and once under the Relaxed condition, with a 5 min rest between tasks. Task durations and intertrial intervals were chosen to balance data quantity and fatigue. Specifically, the measurement periods provided sufficient repetitions to obtain stable neck EMG features, while the rest intervals were long enough to minimize carry-over fatigue across trials.

1) Grip Strength: Participants performed three maximal grips with the right hand, followed by three with the left, using a digital hand dynamometer. I collected two sets of measurements—one for training and one for testing—yielding 12 grips per condition. In the Forceful condition, participants gripped as hard as possible; in the Relaxed condition, they kept force below 50% of their own maximum.

The specific procedure was as follows. Participants measured grip force with the right hand for 5 s, followed by a 5-s rest. This was repeated three times consecutively. They then switched to the left hand and performed the same sequence of 5-s measurement and 5-s rest, repeated three times. This procedure constituted one set. Two sets were conducted in total: one for training data collection and one for test data collection. The task was performed under both the Forceful and Relaxed conditions, resulting in four sets in total.

2) Golf Putter Practice: Participants executed five practice swings with a putter. I collected two sets (training and testing), for ten swings per condition. In the Forceful condition they gripped the club firmly; in the Relaxed condition they gripped lightly.

Participants performed a practice swing once every 3 s, repeating this five times consecutively, followed by a 10-s rest. This procedure constituted one set, and two sets were conducted in total: one for training data collection and one for test data collection. The task was performed under both the Forceful and Relaxed conditions, resulting in four sets in total.

3) Golf Putter: Participants attempted five putts from 2 m. I collected two sets (training and testing), for 10 putts per condition. Force instructions matched those of the practice task.

Participants performed golf putting at a pace of once every 4 s, repeating this five times consecutively, followed by a 10-s rest. This procedure constituted one set, and two sets were conducted in total: one for training data collection and one for test data collection. The task was performed under both the Forceful and Relaxed conditions, resulting in four sets in total.

4) Piano Scale: Participants played a C-major scale across the full 88-key range of a digital piano with the right hand, using self-selected fingering and tempo. I collected two sets (training and testing), for two complete passes per condition. Key-press force was firm in the Forceful condition and soft in the Relaxed condition.

Participants performed an 88-note scale over 20 s, followed by a 10-s rest. This procedure constituted one set, and two sets were conducted in total: one for training data collection and one for test data collection. The task was performed under both the Forceful and Relaxed conditions, resulting in four sets in total.

Note that I excluded the left hand from the piano scale task because, for novices, ascending the C-major scale with the left hand is particularly difficult due to the reverse sequence of finger movements compared to the right hand. Therefore, this task targeted only overexertion of the right arm.

5) Piano Tremolo: Tremolo is a technique in which two notes are rapidly alternated. In the present task, participants used the thumb and little finger to alternately play the notes C and E. Participants performed a 20 s tremolo alternating between C and E, once with the right hand and once with the left. I collected two sets (training and testing), totaling 80 s of tremolo per condition. Force instructions matched those of the scale task.

Participants performed a 20-s tremolo with the right hand, followed by a 10-s rest. They then performed a 20-s tremolo with the left hand, followed by another 10-s rest. This procedure constituted one set, and two sets were conducted in total: one for training data collection and one for test data collection. The task was performed under both the Forceful and Relaxed conditions, resulting in four sets in total.

C. EMG Analysis

Raw EMG was band-pass filtered (20–450 Hz) and notch filtered at 50 Hz to suppress motion artifacts and mains hum [16]. RMS amplitude was computed in 100 ms sliding windows and normalized by the peak rms during MVC, yielding %MVC. Samples with normalized amplitude exceeding 100 %MVC and statistical outliers (interquartile-range criterion) were removed; the resulting gaps were linearly interpolated. For each task, I calculated the mean %MVC over the entire execution window (1–2 min). Arm activity was represented as the average of the right- and left-forearm channels.

To test whether mean %MVC differed between Forceful and Relaxed conditions, I first applied the Shapiro–Wilk test to each paired dataset. Normally distributed pairs were compared using two-tailed paired t-tests; nonnormal pairs were analyzed with Wilcoxon signed-rank tests. All p-values were corrected for multiple comparisons using the Benjamini–Hochberg procedure. Effect sizes were reported as Cohen's d (for t-tests) and r (for Wilcoxon tests) [17].

D. Classification and Regression Models

To build the machine-learning models, I constructed feature vectors as follows.

1) Segmented the neck %MVC time series into 10 s windows.

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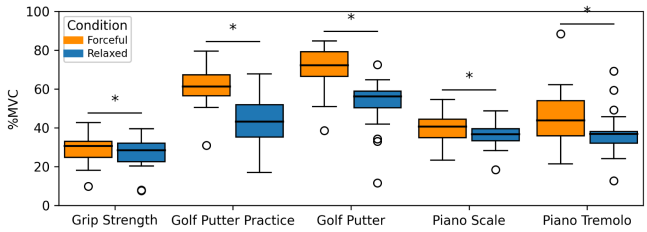

Fig. 1. Boxplots of participantwise mean %MVC of bilateral forearm muscles by task and effort condition. Asterisks denote significant forceful–relaxed differences (BH-adjusted $p < 0.05$).

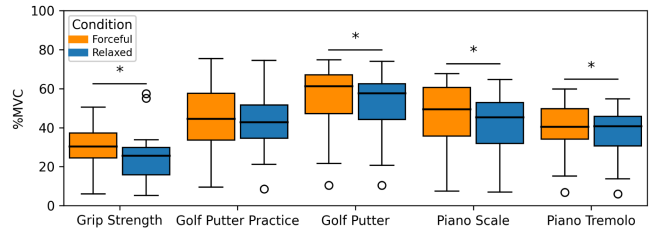

Fig. 2. Boxplots of participantwise mean posterior-neck %MVC by task and effort condition. Asterisks denote significant forceful–relaxed differences (BH-adjusted $p < 0.05$).

2) For each window, computed the mean, standard deviation, and maximum, which are representative temporal features.
3) Slid the window forward by 1 s and repeated the calculations until the end of the recording.

For classification, labels were 1 for the Forceful condition and 0 for the Relaxed condition. For regression, the target was the mean %MVC of both arms within each 10 s window.

In studies of muscle fatigue estimation using EMG, when capturing steady-state changes rather than instantaneous muscle output, analysis is often conducted with a window length of approximately 5–10 s [18], [19]. Similarly, the present study aimed not to detect brief, transient exertion, but to classify whether participants were in a state of exertion during each task. Therefore, I used a 10-s window with a 1-s sliding step.

For the classification task, four models were trained: logistic regression, support vector machine (SVM), XGBoost, and random forest. For the regression task, four models were trained: linear regression, support vector regression (SVR), XGBoost, and random forest. I used Python, and models were implemented using scikit-learn (v1.7.1) and XGBoost (v3.0.2), with default hyperparameters. A separate participant-specific model was trained for each individual.

For each task, I trained on the dataset collected for training and evaluated on the dataset collected for testing. Classification performance was assessed using accuracy and F1 score; regression performance was assessed using root-mean-squared error (RMSE) and the coefficient of determination (R^2).

III. RESULTS

A. Comparison of %MVC Across Tasks and Conditions

Fig. 1 shows boxplots of participantwise mean %MVC across all tasks and both effort conditions. Two-tailed paired t-tests confirmed that the Forceful instruction reliably elevated arm activity relative to Relaxed in every task, indicating successful manipulation of arm tension: Grip Strength ($t(17) = 2.7$, $p_{\text{adj}} = 0.016$, $d = 0.63$), Golf Putter Practice ($t(17) = 6.7$, $p_{\text{adj}} = 1.9 \times 10^{-5}$, $d = 1.6$), Golf Putter ($t(17) = 5.0$, $p_{\text{adj}} = 1.8 \times 10^{-4}$, $d = 1.2$), Piano Scale ($t(17) = 3.0$, $p_{\text{adj}} = 0.0094$, $d = 0.71$), and Piano Tremolo ($t(17) = 5.9$, $p_{\text{adj}} = 4.7 \times 10^{-5}$, $d = 1.4$).

Fig. 2 shows boxplots of participantwise mean %MVC from the posterior-neck EMG sensor. In four of the five tasks, neck activity was significantly higher in the Forceful than in the Relaxed condition: Grip Strength ($W = 31$, $p_{\text{adj}} = 0.029$, $r = 0.56$), Golf Putter ($W = 33$, $p_{\text{adj}} = 0.029$, $r = 0.54$), Piano Scale ($t(17) = 2.5$, $p_{\text{adj}} = 0.029$, $d = 0.59$), and Piano Tremolo ($t(17) = 2.5$, $p_{\text{adj}} = 0.029$, $d = 0.59$). No significant difference was observed for Golf Putter Practice ($t(17) = 1.4$, $p_{\text{adj}} = 0.17$, $d = 0.34$). These results indicate that posterior-neck EMG amplitude increases in concert with upper-limb exertion across multiple tasks.

TABLE 1. Accuracy, F1, and RMSE (%, $M \pm SD$) of the Random Forest

	Grip	Practice	Putter	Scale	Tremolo
Accuracy	84 ± 14	65 ± 19	70 ± 23	83 ± 21	78 ± 22
F-1 score	83 ± 16	60 ± 27	62 ± 31	79 ± 27	76 ± 27
RMSE	7 ± 3	13 ± 6	14 ± 6	8 ± 3	9 ± 4

B. Classification Results

Across models, a consistent pattern emerged: higher accuracies for Grip Strength (78%–84%), Piano Scale (80%–86%), and Piano Tremolo (75%–80%); and lower accuracies for Putter Practice (64%–68%) and Golf Putter (61%–70%). Averaged across tasks, overall accuracies (%, mean ± SD) were: logistic regression 76 ± 22, SVM 74 ± 23, XGBoost 73 ± 22, and random forest 76 ± 21. The corresponding F1 scores (%, mean ± SD) were: logistic regression 71 ± 29, SVM 71 ± 28, XGBoost 68 ± 28, and random forest 72 ± 27. Considering both accuracy and F1, the random forest achieved the best overall classification performance among the four classifiers; Table 1 shows the results for each task using the random forest model.

C. Regression Results

Across models, a consistent pattern emerged: lower RMSE for Grip Strength (7%–8%), Piano Scale (7%–9%), and Piano Tremolo (9%–10%); and higher RMSE for Golf Putter Practice (13%–17%) and Golf Putter (14%–17%). Averaged across tasks, RMSE (%, mean ± SD) was: linear regression 12 ± 8, SVR 10 ± 5, XGBoost 11 ± 6, and random forest 10 ± 5. Considering RMSE, the random forest achieved the best overall regression performance; Table 1 shows the results. The coefficient of determination between predictions and ground truth is $R^2 = 0.72$.

IV. DISCUSSION

Previous ergonomics studies have mainly treated cervical EMG as an indicator of local neck muscle load during light tasks [15]. In contrast, the present study exploits a single neck-mounted EMG sensor as a convenient proxy for bilateral forearm activation and attempts to indirectly estimate upper-limb overexertion.

The intended use of my method is to determine, during movements such as golf and piano, whether the user is tensing; the simplest formulation is a binary classification of tensed versus relaxed. The corresponding results are the accuracies and the F1 scores in Table 1. The accuracy and F1 reached about 80% for less dynamic activities (Grip

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Strength and the piano tasks). By contrast, performance decreased to just under 70% for the more dynamic golf tasks, likely due to motion artifacts caused by neck movement during the swing. Aggregated across tasks, mean accuracy was 76%; given a 50% chance level, this indicates that the binary decision can be made with substantially above-chance accuracy.

For finer-grained estimation of upper-limb exertion, I regress the mean %MVC across both arms. The corresponding evaluation results are the RMSE values in Table 1, approximately 7%–14%. The relationship between predictions and ground truth gives $R^2 = 0.72$, indicating a practically useful level of estimation. Thus, while the method is not suited to precise estimation of the exact %MVC of both arms, it is useful when a coarse, multilevel estimate suffices, in which case the regression outputs can be leveraged effectively.

The experimental results suggest that upper-limb exertion can potentially be estimated using a single EMG sensor placed on the posterior neck. Although errors may increase depending on the task, this unique placement on the posterior neck offers advantages not available in conventional measurement methods. This enables estimation of upper-limb exertion in settings where the arms frequently contact external objects or where visible instrumentation is undesirable for aesthetic reasons.

This work is an exploratory demonstration of estimating upper-limb exertion from a single EMG sensor on the posterior neck; accordingly, the findings should not be overgeneralized. First, validation was conducted on isolated instances of one target movement at a time. In real use, multiple movements typically occur in sequence—for example, golfers alternate between driver shots and putts, and pianists interleave scale passages and tremolo within a piece. Whether performance is maintained throughout continuous, heterogeneous sequences remains to be established. Second, the models were trained per participant using brief calibration data; generalization across users, sessions, and sensor repositioning was not assessed. Third, the sample size (18 participants) may limit external validity. Finally, I evaluated a single neck site (upper trapezius) and one sensing platform; robustness to different placements, devices, and motion artifacts during vigorous head/neck movement (e.g., golf swings) warrants further investigation.

V. CONCLUSION

Unlike previous work that used cervical EMG primarily to quantify local neck muscle load, this study demonstrated the feasibility of using a single EMG sensor on the posterior neck as an indirect marker of upper-limb overexertion. The findings are as follows.

1) Coactivation: Neck muscles coactivate with arm exertion; when the arms are tensed, neck %MVC also increases.
2) Binary classification: Using a single neck-mounted EMG sensor, whether the arms were tensed could be inferred with an average accuracy of 76% and an F1 score of 72%.
3) Regression: When estimating the mean %MVC across both arms, the model achieved an RMSE of 10% and an $R^2 = 0.72$.

This neck placement offers a practical, socially acceptable, and unobtrusive alternative that is unlikely to interfere with natural movement. To verify robustness, I plan to examine how perspiration and neck movement during vigorous exercise may influence estimation accuracy.

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